

Problem 1.7

We argue for interchanging integration and differentiation as in Problem 6:
One has

$$\int_0^1 \exp \left(i \sum_{k=1}^n a_k r_k(\omega) \right) d\omega = 2^{-n} \sum \exp \left(i \sum_{k=1}^n a_k \pm 1 \right),$$

where the sum extends over all n -long sequences of 1 and -1 (see P.1.6 in Billingsley for notation). This immediately yields by the discussion above and polarization identity

$$\int_0^1 \exp \left(i \sum_{k=1}^n a_k r_k(\omega) \right) d\omega = \prod_{k=1}^n \frac{e^{ia_k} + e^{-ia_k}}{2} = \prod_{k=1}^n \cos a_k.$$

Now choosing $a_k = t2^{-k}$ and using the hint yields

$$\begin{aligned} \prod_{k=1}^n \cos \frac{t}{2^k} &= \int_0^1 \exp \left(it \sum_{k=1}^n 2^{-k} r_k(\omega) \right) d\omega \\ &= \int_0^1 \underbrace{\exp \left(it \sum_{k=1}^n 2^{-k} r_k(\omega) \right) - \exp(it(2\omega - 1))}_{\rightarrow 0, \text{ uniformly } (*) \text{ in } \omega \text{ as } n \rightarrow \infty} d\omega + \int_0^1 \exp(it(2\omega - 1)) d\omega \\ &\xrightarrow{n \rightarrow \infty} \int_0^1 \exp(it(2\omega - 1)) d\omega = \frac{\sin t}{t}. \end{aligned}$$

This yields

$$\prod_{k=1}^{\infty} \cos \frac{t}{2^k} = \frac{\sin t}{t}$$

Now putting $t = \frac{\pi}{2}$ yields for the RHS $\frac{2}{\pi}$. For the LHS We use $\cos(\pi/4) = \sqrt{2}/2$ (can be deduced from first principles in analysis) and then iteratively use for $k \geq 2$ $\cos \frac{\pi}{2^k} = \frac{1}{\sqrt{2}} \sqrt{\cos \left(\frac{\pi}{2^{k-1}} \right) + 1}$ which the yields Vieta's formula.

Remark ()*: We do not use any integral convergence theorems, simply monotonicity of the integral and properties of the dyadic expansion, work out the details.